LabLedger Database Structure

Print reference for the MongoDB collections, fields, relationships, validation rules, and indexes defined by the current Mongoose models.

At a glance

Collection	Purpose	Relationship
items	Inventory records and item-level metadata.	Parent record; one item can have many unit records and history entries.
repairments	One record per tracked physical unit, including condition and repair details.	repairments.item_id references items._id.
history	Audit entries describing changes to an item or a unit repairment.	history.item_id references items._id; repairment_id optionally references repairments._id.

Model notes

This is a schema reference, not a live database dump. MongoDB stores flexible BSON documents; fields may be absent when optional and defaults apply on Mongoose writes. ObjectId references are application-level references. Embedded location, image, update, spare-part, and change records are stored inside their parent documents, not as separate collections.

The source models define collections explicitly as items, repairments, and history. Repairment records represent individual units; the parent item stores aggregate quantities. Golden is a unit status and is used for functional units with no repairment history.

Collection 1 Items

Collection: **items** | Mongoose model: **Item** | timestamps: **dates.created**, **dates.modified**

Field	Type	Rules and meaning
_id	ObjectId	MongoDB document identifier; generated by Mongoose.
name	String	Required; trimmed.
part_num	String	Optional; trimmed.
type	String	Optional; allowed values: pcb, module, else.
stored	Boolean	Required. Determines whether the item is in lab inventory or delivered.
deleted	Boolean	Default false; soft-delete flag.
location	Embedded object	Optional overall; warehouse, section, pack. Warehouse is required for stored items, except the under-repair lab location case.
location.warehouse	String	Trimmed. Storage area; required when stored, subject to under-repair exception.
location.section	String	Trimmed; optional storage subsection.
location.pack	String	Trimmed; optional pack/bin.
owner	String	Optional; trimmed.
organization	String	Optional; trimmed.
serial_num	Array	Default empty array; serial numbers associated with units.
under_repairment	Boolean	Default false; marks parent item as having units under repair.
total_quantity	Integer	Default 1; minimum 1.
available_quantity	Integer	Default 1, initialized to total_quantity for new records; minimum 0 and cannot exceed total_quantity.
description	String	Optional; trimmed.
tags	Array	Default empty array.
updates	Array	Default empty array. Each update has required trimmed text and createdAt date (default now).
images	Array	Default empty array. Each image has required path, required originalName, and uploadedAt date (default now).
delivered_by	String	Optional; trimmed.
delivered_to	String	Trimmed; must be present if stored is false and empty if stored is true.
dates.created	Date	Created timestamp managed by Mongoose.
dates.modified	Date	Updated timestamp managed by Mongoose.
edit_count	Number	Default 0; minimum 0.

Indexes

{ deleted: 1, "dates.created": -1 } supports deleted-state and newest-created sorting. **{ part_num: 1 }** indexes part number.

Embedded item subdocuments

location has no independent _id. Each image and update has a subdocument _id by Mongoose defaults. Image fields: path, originalName, uploadedAt. Update fields: text, createdAt.

Collection 2 Repairments

Collection: **repairments** | Mongoose model: **Repairment** | timestamps: **createdAt**, **updatedAt**

Field	Type	Rules and meaning
_id	ObjectId	MongoDB document identifier; generated by Mongoose.
item_id	ObjectId reference	Required; references Item in the items collection.
serial_num	String	Optional; trimmed; identifies this individual unit.
status	String	Required. Allowed: golden, repaired, unrepairable, repairing, awaiting_spare_part, delivered.
delivered_to	String	Optional; trimmed. Required by service logic when status is delivered.
delivered_by	String	Optional; trimmed.
delivered_at	Date	Optional delivery date/time.
field_test_date	Array	Default empty array.
repairer	Array	Default empty array; names of repairers.
spare_part	Array	Default empty array. Each entry has required trimmed part and required price >= 0; subdocument _id disabled.
updates	Array	Default empty array. Each update has required trimmed text and createdAt date (default now).
deleted	Boolean	Default false; soft-delete flag.
createdAt	Date	Created timestamp managed by Mongoose.
updatedAt	Date	Updated timestamp managed by Mongoose.

Indexes

{ item_id: 1, deleted: 1, createdAt: 1 } supports listing unit records by parent item and deleted state in creation order.

Unit status meanings

golden: functional and never repaired; **repairing** and **awaiting_spare_part**: active repair states; **repaired** and **unrepairable**: repair outcome states; **delivered**: unit delivered from inventory.

Collection 3 History

Collection: **history** | Mongoose model: **History** | no automatic timestamps; uses a date field.

Field	Type	Rules and meaning
_id	ObjectId	MongoDB document identifier; generated by Mongoose.
item_id	ObjectId reference	Required; references Item in the items collection.
repairment_id	ObjectId reference or null	Optional; references Repairment. Defaults to null.
fields	Array	Each entry records field_name (required), from (Mixed; defaults null), and to (Mixed).
date	Date	Defaults to current date/time.

Indexes

{ item_id: 1, date: 1 } supports retrieving an item history chronologically.

Relationship map

From	Relationship	To
items._id	one to many via repairments.item_id	repairments.item_id
items._id	one to many via history.item_id	history.item_id
repairments._id	optional reference via history.repairment_id	history.repairment_id

Operational notes

Database: MongoDB through Mongoose. The sample connection uses a replica set because transactional operations require replica-set support. The model definitions do not define unique indexes. Soft deletion is represented by deleted flags on items and repairments; history has no deleted flag.

Source: current schema definitions in src/models/item.model.js, src/models/repairment.model.js, and src/models/history.model.js. This reference describes schema rules; runtime service logic may enforce additional workflow requirements.